

INTRODUCTION

Industrial Facility Design

FACILITIES

- Defined as buildings / place where people, material, and machines come together for a stated purpose – typically to make a tangible product or provide a service. The purpose of Product or Service are
- Lower cost,
- Higher quality,
- Using the least amount of resources.

- **Importance of Facilities Planning & Design:**
- Manufacturing and Service companies spend a significant amount of time and money to design or redesign their facilities. This is an extremely important issue and must be addressed before products are produced or services are rendered.
A poor facility design can be costly and may result in:
 - poor quality products,
 - low employee morale,
 - customer dissatisfaction.

- **Disciplines involved in Facilities Planning:**

Within the engineering profession:

- Civil engineers,
- Electrical engineers,
- Industrial engineers,
- Mechanical engineers.

Additionally,

- Architects,
- Consultants,
- General contractors,
- Managers,
- Real estate brokers, and
- Urban planners

- Facility Planning and Design tools vary from checklists, cookbook type approaches to highly sophisticated mathematical modeling approaches.
- A practical approach to facilities planning will be employed taking advantage of empirical and analytical approaches using both tradi. **Applications of**

Facilities Planning (FP):

Facilities Planning (FP) can be applied to planning of:

- A new hospital,
- **An assembly department,**
- **An existing warehouse,**
- The baggage department in an airport,
- Department building of IE in UMT
- **A production plant,**
- A retail store,
- A dormitory,
- A bank,
- An office,
- A cinema,
- A parking lot,
- Or any portion of these activities etc...

- Facilities Planning & Design determines how an activities tangible fixed assets best support achieving the activity's objectives. i.e.
- What is the objective of the facility?
- How the facility achieves that objective?
- **In the case of a manufacturing firm:**
Facilities Planning & Design involves the determination of how the manufacturing facility best supports.
- Facilities Planner considers the facility as a dynamic entity. Therefore continuous improvement is an integral element

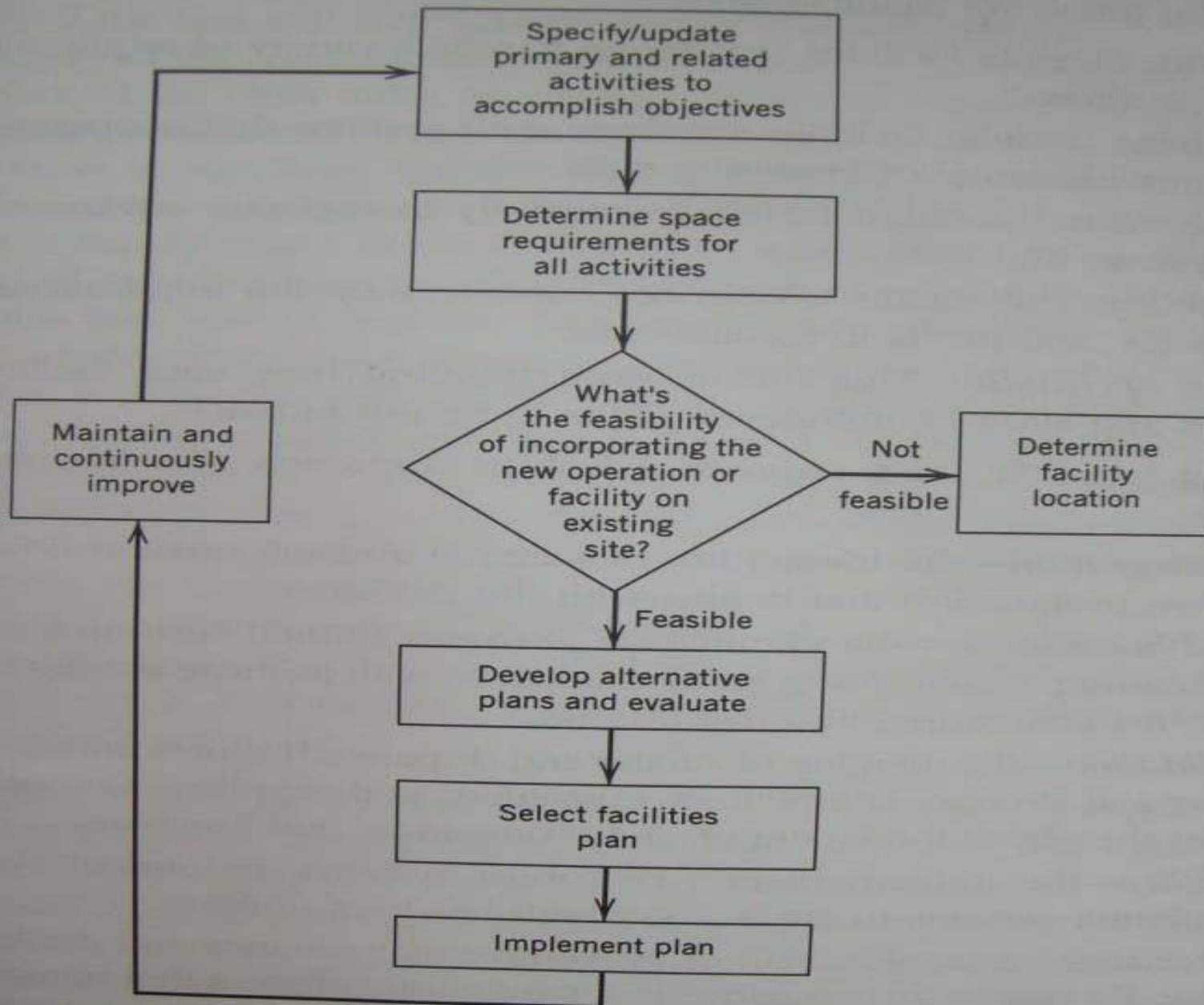
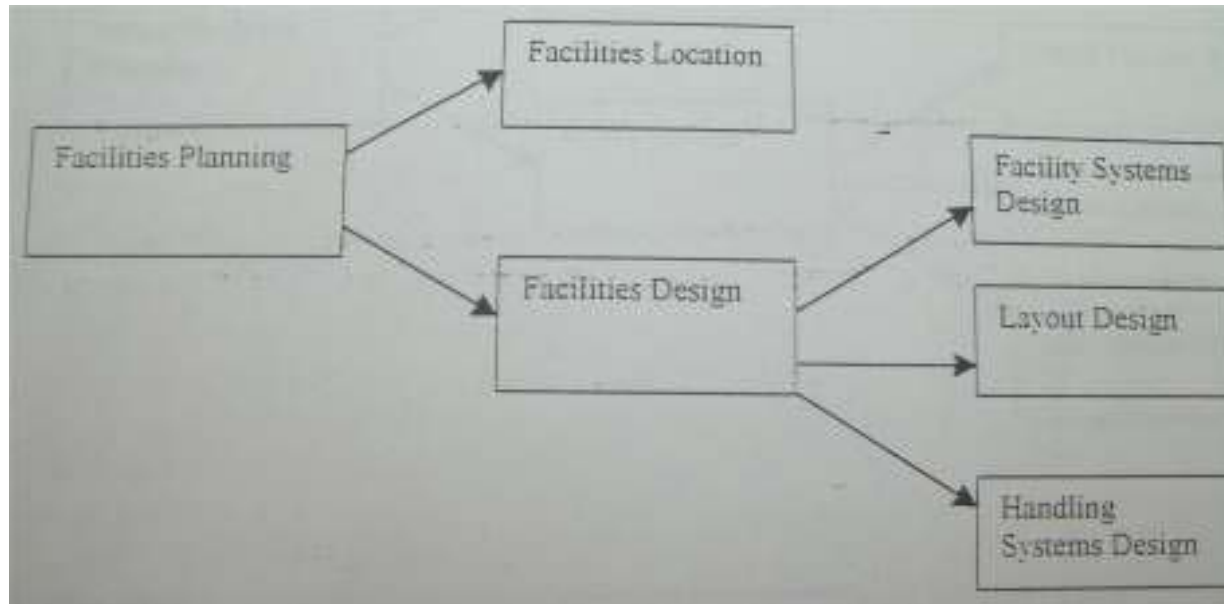


Figure 1.1 Continuous improvement facilities planning cycle.

Facilities Planning $\neq$ Facility Location
Facilities Design
Facilities Layout
Plant Layout

Facilities Planning Hierarchy:



- **Facilities Location (Macro Aspect of FP):**

Location of the facility refers to its placement with respect to customers, suppliers, and other facilities with which it interfaces.

- **Facilities Design (Micro Aspect of FP):**

Design components of a facility consists of

- **Facilities Systems:**

Consists of the structural systems, the atmospheric systems, the lighting/electricity/communication systems, the life safety systems and the sanitation systems.

- **Layout:**

Consists of all equipment, machinery and furnishings within the building.

- **Handling Systems:**

Consists of the mechanism need to satisfy the required facility interactions.

- **For a manufacturing system:**
- **Facility Systems** – the structure (of building), power, light, gas, heat, ventilation, air-conditioning, water and sewage needs.
- **Layout** – the production areas, related support areas, personnel areas.
- **Handling Systems** – the materials- personnel, information, and equipment to support manufacturing.

1.2. Significance of Facilities Planning

To understand the significance of Facilities Planning (FP) consider the following questions:

- What impact does facilities planning have on handling and maintenance cost?
- What impact does facilities planning have on employee morale, and how does employee morale impact operating costs?
- In what do organizations invest the majority of their capital, and how convertible is their capital once invested?
- What impact does facilities planning have on the management of a facility?
- What impact does facilities planning have on facility's capability to adapt to change and satisfy future requirements?

1.3. Objectives of Facilities Planning

Objectives of FP is to plan a facility that achieves both facilities location and design objectives.

1.3.1. Objectives of Industrial Facility Location:

Objective of Industrial Facility Location is to determine the location which, in consideration of all factors affecting deliver-to-customers cost of the products to be manufactured, will be minimized.

1.3.2. Some Typical Facilities Design Objectives are to:

1. Support the organization's vision through improved material handling, material control, and good housekeeping.
2. Effectively utilize people, equipment, space and energy.
3. Minimize capital investment.
4. Be adaptable and promote ease of maintenance.
5. Provide for employee safety and job satisfaction.